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Transducer surround having a barometric vent

Abstract

A transducer assembly comprising: a frame positioned within an interior chamber defined by an enclosure comprising a port to an ambient environment; a surround comprising a fixed portion sealed to the frame and a movable portion dividing the interior chamber into a front volume chamber coupling a first side of the surround to the port and a back volume chamber coupled to a second side of the surround; and a barometric vent coupled to the fixed portion to vent a pressure between the front volume chamber and the back volume chamber.

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Background/Summary

FIELD

(1) An aspect of the disclosure is directed to a transducer surround having a barometric vent. Other aspects are also described and claimed.

BACKGROUND

(2) Portable communications or listening devices (e.g., smart phones, earphones, etc.) have within them one or more transducers that convert an input electrical audio signal into a sound pressure wave output that can be heard by the user, or a sound pressure wave input into an electrical audio signal. The transducer (e.g., a speaker) can be used to, for example, output sound pressure waves corresponding to the voice of a far end user, such as during a telephone call, or to output sound pressure waves corresponding to sounds associated with a game or music the user wishes to play. Due to the relatively low profile of the portable devices, the transducers also have a relatively low profile, which in turn, can make it difficult to maintain optimal sound quality.

SUMMARY

(3) An aspect of the disclosure is directed to a device and/or transducer assembly having a barometric vent formed through the surround to reduce the device footprint required for the barometric vent and increase interior volumes for improved acoustic performance.

Representatively, in some aspects, a barometric vent is used to equalize pressure between chambers within the transducer (e.g., front volume chamber and back volume chamber). To accommodate the barometric vent, the transducer enclosure or frame footprint is often increased to include an additional pocket or structure that forms an opening between the chambers and a membrane sealed over the opening to form the barometric vent. This increased footprint, however, may take away from an acoustic volume within the enclosure and reduce the size of the interior chambers (e.g., reducing the back volume chamber). This, in turn, may reduce the acoustic performance of the

transducer. The instant disclosure, on the other hand, provides for a barometric vent through the surround so that an increase in the frame footprint is not necessary. Representatively, in one aspect, the surround may include a projecting portion sealed to the frame and the barometric vent may be stacked on top of the projecting portion. Since the barometric vent is stacked on top of the surround-to-frame seal a smaller footprint is achieved. The smaller footprint means that there is more volume and X-Y space for other features, such as additional back volume, or additional space for a harp mesh, which provides low frequency performance benefits.

(4) Representatively, in some aspects, the disclosure is directed to a transducer assembly comprising: a frame positioned within an interior chamber defined by an enclosure comprising a port to an ambient environment; a surround comprising a fixed portion sealed to the frame and a movable portion dividing the interior chamber into a front volume chamber coupling a first side of the surround to the port and a back volume chamber coupled to a second side of the surround; and a barometric vent coupled to the fixed portion to vent a pressure between the front volume chamber and the back volume chamber. In some aspects, the fixed portion may include a projection extending radially outward from the movable portion and the barometric vent is coupled to the projection. The movable portion may include a rectangular shape and the projection extends radially outward from a corner of the rectangular shape. In other aspects, the movable portion may include a hexagonal shape and the projection extends radially outward from a side of the hexagonal shape. The projection may extend radially outward from less than an entire perimeter of the movable portion. In some aspects, the barometric vent may include an air permeable membrane covering an opening through the fixed portion of the surround and the frame, and a perimeter of the air permeable membrane is sealed to the fixed portion. In some aspects, the perimeter of the air permeable membrane is sealed to a top side of the fixed portion of the surround, and a bottom side of the fixed portion of the surround directly below the perimeter is sealed to the frame. In some aspects, the barometric vent includes a triangular shape, a rectangular shape, a square shape, an elliptical shape or a racetrack shape. In still further aspects, the movable portion may include a rolled portion that surrounds a piston portion that is operable to vibrate to generate sound. In other aspects, the movable portion is symmetrical about two axes.

(5) In still further aspects, a portable electronic device is disclosed including an enclosure having a top cover and a bottom cover that define an interior chamber and an acoustic port to the interior chamber; a frame positioned within the interior chamber and sealed to the top cover or the bottom cover; a surround comprising a perimeter portion sealed to the frame and a movable portion operable to generate sound that divides the interior chamber into a front volume chamber coupling a first side of the movable portion to the acoustic port and a back volume chamber coupled to a second side of the movable portion; and a barometric vent comprising an air permeable membrane sealed to the perimeter portion to vent a pressure between the front volume chamber and the back volume chamber. In some aspects, the perimeter portion includes a projection extending radially outward from the movable portion and the barometric vent is coupled to the projection. In still further aspects, the movable portion includes a rectangular shape and the projection extends radially outward from a corner of the rectangular shape. In other aspects, the movable portion may include a hexagonal shape and the projection extends radially outward from a side of the hexagonal shape. The projection may extend radially outward from less than an entire perimeter of the movable portion. In some aspects, the barometric vent includes an air permeable membrane covering an opening through the perimeter portion of the surround and the frame, and a perimeter of the air permeable membrane is sealed to the perimeter portion. In some aspects, the perimeter of the air permeable membrane is sealed to a top side of the perimeter portion of the surround, and a bottom side of the fixed portion of the surround directly below the perimeter of the surround is sealed to the frame. In some aspects, the barometric vent includes triangular shape. In still further aspects, the movable portion comprises a rolled portion that surrounds a piston portion that is operable to vibrate to generate sound. The movable portion may further be symmetrical about two

axes.

(6) The above summary does not include an exhaustive list of all aspects of the present disclosure. It is contemplated that the disclosure includes all systems and methods that can be practiced from all suitable combinations of the various aspects summarized above, as well as those disclosed in the Detailed Description below and particularly pointed out in the claims filed with the application. Such combinations have particular advantages not specifically recited in the above summary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The aspects are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings in which like references indicate similar elements. It should be noted that references to “an” or “one” aspect in this disclosure are not necessarily to the same aspect, and they mean at least one.

(2) FIG. 1 illustrates a cross-sectional side view of one aspect of a portable electronic device and/or transducer assembly having a barometric vent.

(3) FIG. 2 illustrates a magnified cross-sectional side view of a portion of the barometric vent of the portable electronic device and/or transducer assembly of FIG. 1.

(4) FIG. 3 illustrates a top view of one representative configuration of the barometric vent of the portable electronic device and/or transducer assembly of FIG. 1.

(5) FIG. 4 illustrates a top view of another representative configuration of the barometric vent of the portable electronic device and/or transducer assembly of FIG. 1.

(6) FIG. 5 illustrates a block diagram of one aspect of a portable electronic device within which a transducer assembly of FIG. 1-FIG. 4 may be implemented.

DETAILED DESCRIPTION

(7) In this section we shall explain several preferred aspects of this disclosure with reference to the appended drawings. Whenever the shapes, relative positions and other aspects of the parts described are not clearly defined, the scope of the disclosure is not limited only to the parts shown, which are meant merely for the purpose of illustration. Also, while numerous details are set forth, it is understood that some aspects of the disclosure may be practiced without these details. In other instances, well-known structures and techniques have not been shown in detail so as not to obscure the understanding of this description.

(8) The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. Spatially relative terms, such as “beneath”, “below”, “lower”, “above”, “upper”, and the like may be used herein for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(9) As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising” specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

(10) The terms “or” and “and/or” as used herein are to be interpreted as inclusive or meaning any

one or any combination. Therefore, “A, B or C” or “A, B and/or C” mean “any of the following: A; B; C; A and B; A and C; B and C; A, B and C.” An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.

(11) FIG. 1 illustrates a cross-sectional side view of one aspect of a transducer assembly that may be integrated into a portable electronic device. Assembly **100** may include a housing, casing or outer enclosure **102** that defines or closes off a chamber in which the constituent electronic components of assembly **100** are contained. In some aspects, it is contemplated that assembly **100** may be a transducer module or a portable or mobile communications device within which the module is integrated, an in-ear device, portable time piece or any other device within which a transducer may be implemented. In some aspects, enclosure **102** may include a top cover **102A** coupled to a bottom cover **102B** that separate a surrounding environment from an encased space or interior chamber **104** formed within enclosure **102**. In some cases, enclosure **102** completely isolates or seals the entire, or a portion of, interior chamber **104** from the surrounding environment. For example, enclosure **102** may form a waterproof or acoustically isolated portion of interior chamber **104** which is impermeable to water and/or air. The interior chamber **104** may be of a sufficient volume and/or size to accommodate the constituent components of assembly **100**. In some aspects, enclosure **102** may also include one or more of an acoustic port **106**. The acoustic port **106** may be, for example, a sound output port through which sound from a transducer such as a speaker positioned within interior chamber **104** may be output. In other aspects, where a microphone is positioned near enclosure acoustic port **106**, it could be a sound input port to allow for input of sound to the microphone.

(12) Representatively, in one aspect shown in FIG. 1, enclosure acoustic port **106** is an acoustic port that is acoustically open to a transducer **108** positioned within interior chamber **104**. In some aspects, transducer **108** may be any type of electroacoustic transducer capable of converting an electrical audio signal into a sound or a sound into an electrical audio signal. Representatively, transducer **108** may be a speaker or a micro-speaker, for example, a miniaturized version of a loudspeaker that uses a moving coil motor to drive sound output. For example, transducer **108** may include a magnet assembly **110** mounted to bottom cover **102B**. A surround **112** having a stiffener **114** coupled to a top side and a voice coil **116** coupled to a bottom side may be positioned over magnet assembly **110**. A perimeter of surround **112** may be sealed to frame **118** and the remainder of surround **112** may be movable such that the stiffener **114** and voice coil **116** may vibrate relative to magnet assembly **110** and output sound. Thus, in some aspects, transducer **108** may be referred to herein as a micro-speaker. In other aspects, where transducer **108** converts sound into an electrical audio signal, it may further be referred to herein as a microphone.

(13) In some aspects, surround **112** may further be considered to divide interior chamber **104** into a front volume chamber **104A** and a back volume chamber **104B** around transducer **108**. In the case where transducer **108** is a speaker, front volume chamber **104A** may form a chamber having a first volume around the sound output face or top side of surround **112**. The front volume chamber **104A** (and first volume) may be considered acoustically coupled to, or otherwise open to, acoustic port **106**. In this aspect, sound pressure waves output from the top side of surround **112** of transducer **108** may pass through front volume chamber **104A** and out to the surrounding ambient environment **120** through acoustic port **106**. Back volume chamber **104B** may have a second volume and surround the back side of surround **112**. Typically, the front volume chamber **104A** is substantially sealed from the back volume chamber **104B**. It is recognized, however, that in some aspects a size, volume, pressure or other aspects of front volume chamber **104A** and/or back volume chamber **104B** may impact the acoustic performance of transducer **108**. Thus, modifying the size, volume and/or pressure of front volume chamber **104A** and/or back volume chamber **104B** may be used to tune the acoustic performance of transducer **108**. For example, in some cases, it may be desirable to increase a size of the back volume chamber **104B** while also equalizing a

pressure between front volume chamber **104A** and back volume chamber **104B**.

(14) With this in mind, a barometric vent **122** between the front and back volume chambers **104A-B** may further be provided to help equalize the pressure between the two chambers. In addition, barometric vent **122** may have a particular arrangement and/or configuration that also increases a size of back volume chamber **104B**. For example, it is contemplated that in some aspects, barometric vent **122** may be arranged within a footprint of surround **112** and frame **118**, instead of outside of surround/frame, such that the additional volume **104C** may be part of back volume **104B** resulting in an increased back volume. For example, in some aspects, additional volume **104C** may be a volume including an acoustically adsorbent material such as a zeolite which may increase low frequency performance when coupled to the back volume **104B**.

(15) Representatively, barometric vent **122** will now be described in more detail in reference to FIG. 2. FIG. 2 illustrates a magnified view of barometric vent **122** of FIG. 1. From this view, it can be understood that surround **112** includes a movable portion **112B** having a stiffener **114** attached thereto that moves or vibrates up and down to output sound as previously discussed. A rolled portion **112A** is further positioned radially outward from, or otherwise around, the movable portion **112B** to allow for the vibration or up/down movement of movable portion **112B**. As can be seen from this view, rolled portion **112A** has a curved or otherwise out-of-plane shape that allows the movable portion **112B** to move up/down in a piston like fashion relative to frame **118**. Extending from rolled portion **112A** is a perimeter portion **112D** of surround **112** that is sealed to a top surface of frame **118** by a frame seal **208** (e.g., a glue or other sealing material). Perimeter portion **112D** may therefore be understood as a fixed portion of surround **112** that connects and suspends surround **112** from frame **118**. Perimeter portion **112D** may extend entirely around rolled portion **112A**. Extending laterally outward from perimeter portion **112D** is a protrusion or extension portion **112C**, which is also sealed to frame **118** and includes barometric vent **122**. In this aspect, protrusion or extension portion **112C** and frame **118** include an opening **202** that provides a vent or pathway between front volume **104A** and back volume **104B**. An air permeable and water-proof membrane **204** is then positioned over opening **202** and sealed to the top side of perimeter portion **112D** of surround **112**.

(16) Representatively, in some aspects, membrane **204** may be sealed to a top side of surround perimeter portion **112D** using a surround seal **206** (e.g., a glue or other sealing material) that is considered stacked directly above the frame seal **208** along the bottom side of surround portion **112D**. Membrane **204** may allow for the passage of air to equalize the pressure between front and back volumes **104A** and **104B**, while preventing the passage of water between the volumes. Barometric vent **122** may be considered formed by a stack up of frame **118**, frame seal **208** sealing surround protrusion **112C** to frame **118**, and surround seal **206** sealing membrane **204** to surround protrusion **112C**. In this aspect, barometric vent **122** may be considered to be within a footprint of frame **118** and does not require or otherwise occupy additional space within front and/or rear volumes **104A-B**. Frame **210** may then be sealed to top cover **202A** by a further sealing layer **210** (e.g., a glue or other sealing material). It should further be understood that the various sealing layers **206**, **208**, **210** may extend entirely around the associated components to seal them together and prevent passage of air and/or water therethrough. It should further be understood that surround **112** is a single integrally formed structure including each of the portions **112A-D** as previously discussed. In addition, protrusion or extension portion **112C** may be understood as extending from only a portion of perimeter portion **112D**. In other words, protrusion or extension portion **112C** does not extend entirely around protrusion or extension portion **112C**.

(17) Representatively, referring now in more detail to surround **112**, as can be seen from FIG. 3, in some aspects, surround **112** may have a substantially rectangular shape with the protrusion or extension portion **112C** extending from only a corner of the rectangular shape. For example, stiffener **114** may be attached to the center movable portion of surround **112** as previously discussed and have a rectangular shape. The rolled portion **112A** extends radially outward from,

and entirely around, the center movable portion and stiffener **114**. Rolled portion **112A** may further be considered to have a substantially rectangular shape (e.g., with rounded corners). Perimeter portion **112D** then extends radially outward from and around rolled portion **112A**, and is attached to frame **118** to suspend surround **112** over the previously discussed vent opening **202** formed through frame **118**. Protrusion portion **112C** then extends radially outward from one of the corners of surround **112** as shown. Barometric vent **122** is formed through protrusion portion **112C**. Representatively, vent opening **202** is formed through protrusion portion **112C** and the portion of frame **118** underlying this protrusion portion **112C**. Membrane **204** is then positioned over opening **202** and sealed around opening **202** to protrusion portion **112C**. As illustrated by this configuration, in some aspects, the opening and/or membrane **204** may have an elongated, elliptical, racetrack like and/or rectangular shape. In this aspect, barometric vent **122** is considered within a footprint of frame **118** such that the remaining volumes and spaces of enclosure can be used to form the back and/or front volume chambers **104A-B** as previously discussed. In addition, it should further be understood that while protrusion portion **112C** is shown extending from a corner of surround **112**, protrusion portion **112C** may extend from another portion of the perimeter of surround **112**. The most important aspect is that protrusion portion **112C** be within the sealed perimeter portion **112D** and near the rolled portion **112A** of surround **112** such that it remains within a footprint of frame **118** so that barometric vent **122** does not occupy additional frame space that could otherwise be used for the back or front volume chambers. For example, in some aspects, protrusion portion **112C** could extend laterally outward from a side of surround **112** such that barometric vent **122** is formed along a side of surround **112**.

(18) FIG. 4 illustrates a top plan view of another representative surround configuration. Similar to the previously discussed surround **112** configurations, surround **112** may have a movable portion **112B** surrounded by a rolled portion **112A** and a perimeter portion **112D** sealed to the underlying frame **118**. In this configuration, however, the movable surround portions **112A** and **112B** define a substantially hexagonal or six-sided surround shape. The movable surround portions **112A** and **112B** may further be understood as defining a shape that is symmetrical about two axes **402**, **404**. It is noted that the movable portion of surround **112** should be symmetrical about two axes **402**, **404** so that the movement is a pistonic like motion up and down and parallel to a vertical axis with little to no rocking and/or rolling. In addition, in this configuration, the extension or protrusion portion **112C** may extend from a side instead of a corner of the hexagonal shaped surround **112**. For example, extension or protrusion portion **112C** may have an almost triangular shape and barometric vent **204** may be a triangular shaped vent formed through portion **112C** as shown. It should be understood, however, that barometric vent **204** may have any shape and/or size so long as it remains within the footprint of frame **118** and surround **112** as discussed, and provides the desired pressure equalization between front and back volumes **104A-B**. For example, barometric vent **204** could have a rectangular, square, triangular, elliptical, a racetrack, or any other shape as desired.

(19) FIG. 5 illustrates a block diagram of one aspect of an electronic device within which the previously discussed transducer including a barometric vent may be implemented. As shown in FIG. 5, device **500** may be any type of portable device within which a transducer and/or vent disclosed herein may be desired, for example, an earpiece (e.g., in-ear earpiece, hearing aid or the like), mobile phone, personal digital assistant, portable timepiece or other portable device. Device **500** may include storage **502**. Storage **502** may include one or more different types of storage such as hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory), volatile memory (e.g., battery-based static or dynamic random-access-memory), etc.

(20) Processing circuitry **504** may be used to control the operation of device **500**. Processing circuitry **504** may be based on a processor such as a microprocessor and other suitable integrated circuits. With one suitable arrangement, processing circuitry **504** and storage **502** are used to run software on device **500**, such as internet browsing applications, voice-over-internet-protocol

(VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. Processing circuitry **504** and storage **502** may be used in implementing suitable communications protocols. Communications protocols that may be implemented using processing circuitry **504** and storage **502** include internet protocols, wireless local area network protocols (e.g., IEEE 802.11 protocols—sometimes referred to as Wi-Fi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol, protocols for handling 3G or 4G communications services (e.g., using wide band code division multiple access techniques), 2G cellular telephone communications protocols, etc.

(21) To minimize power consumption, processing circuitry **504** may include power management circuitry to implement power management functions. For example, processing circuitry **504** may be used to adjust the gain settings of amplifiers (e.g., radio-frequency power amplifier circuitry) on device **500**. Processing circuitry **504** may also be used to adjust the power supply voltages that are provided to portions of the circuitry on device **500**. For example, higher direct-current (DC) power supply voltages may be supplied to active circuits and lower DC power supply voltages may be supplied to circuits that are less active or that are inactive. If desired, processing circuitry **504** may be used to implement a control scheme in which the power amplifier circuitry is adjusted to accommodate transmission power level requests received from a wireless network.

(22) Input-output devices **506** may be used to allow data to be supplied to device **500** and to allow data to be provided from device **500** to external devices. Display screens, microphone acoustic ports, speaker acoustic ports, and docking ports are examples of input-output devices **506**. For example, input-output devices **506** can include user input-output devices **508** such as buttons, touch screens, joysticks, click wheels, scrolling wheels, touch pads, keypads, keyboards, microphones, cameras, etc. A user can control the operation of device **500** by supplying commands through user input devices **508**. Display and audio devices **510** may include liquid-crystal display (LCD) screens or other screens, light-emitting diodes (LEDs), and other components that present visual information and status data. Display and audio devices **510** may also include audio equipment such as speakers and other devices for creating sound. Display and audio devices **510** may contain audio-video interface equipment such as jacks and other connectors for external headphones and monitors.

(23) Wireless communications devices **512** may include communications circuitry such as radiofrequency (RF) transceiver circuitry formed from one or more integrated circuits, power amplifier circuitry, passive RF components, antennas, and other circuitry for handling RF wireless signals. Wireless signals can also be sent using light (e.g., using infrared communications). Representatively, in the case of a speaker acoustic port, the speaker may be associated with the port and be in communication with an RF antenna for transmission of signals from the far end user to the speaker.

(24) Returning to FIG. 5, device **500** can communicate with external devices such as accessories **514**, computing equipment **516**, and wireless network **518** as shown by paths **520** and **522**. Paths **520** may include wired and wireless paths. Path **522** may be a wireless path. Accessories **514** may include headphones (e.g., a wireless cellular headset or audio headphones) and audio-video equipment (e.g., wireless speakers, a game controller, or other equipment that receives and plays audio and video content), a peripheral such as a wireless printer or camera, etc.

(25) Computing equipment **516** may be any suitable computer. With one suitable arrangement, computing equipment **516** is a computer that has an associated wireless access point (router) or an internal or external wireless card that establishes a wireless connection with device **500**. The computer may be a server (e.g., an internet server), a local area network computer with or without internet access, a user's own personal computer, a peer device (e.g., another portable electronic device), or any other suitable computing equipment.

(26) Wireless network **518** may include any suitable network equipment, such as cellular telephone base stations, cellular towers, wireless data networks, computers associated with wireless networks,

etc. For example, wireless network **518** may include network management equipment that monitors the wireless signal strength of the wireless handsets (cellular telephones, handheld computing devices, etc.) that are in communication with network **518**.

(27) While certain aspects have been described and shown in the accompanying drawings, it is to be understood that such aspects are merely illustrative of and not restrictive on the broad disclosure, and that the disclosure is not limited to the specific constructions and arrangements shown and described, since various other modifications may occur to those of ordinary skill in the art. The description is thus to be regarded as illustrative instead of limiting. For example, although a speaker is specifically disclosed herein, the vent disclosed herein could be used with other types of transducers, for example, microphones. Still further, although a portable electronic device such as a mobile communications device is described herein, any of the previously discussed valve and transducer configurations may be implemented within a tablet computer, personal computer, laptop computer, notebook computer, headphones and the like. In addition, to aid the Patent Office and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants wish to note that they do not intend any of the appended claims or claim elements to invoke 35 U.S.C. 112(f) unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A transducer assembly comprising: a frame positioned within an interior chamber defined by an enclosure comprising a port to an ambient environment; a surround comprising a fixed portion sealed to the frame and a movable portion dividing the interior chamber into a front volume chamber coupling a first side of the surround to the port and a back volume chamber coupled to a second side of the surround; and a barometric vent coupled to the fixed portion to vent a pressure between the front volume chamber and the back volume chamber.
2. The transducer assembly of claim 1 wherein the fixed portion comprises a projection extending radially outward from the movable portion and the barometric vent is coupled to the projection.
3. The transducer assembly of claim 2 wherein the movable portion comprises a rectangular shape and the projection extends radially outward from a corner of the rectangular shape.
4. The transducer assembly of claim 2 wherein the movable portion comprises a hexagonal shape and the projection extends radially outward from a side of the hexagonal shape.
5. The transducer assembly of claim 2 wherein the projection extends radially outward from less than an entire perimeter of the movable portion.
6. The transducer assembly of claim 1 wherein the barometric vent comprises an air permeable membrane covering an opening through the fixed portion of the surround and the frame, and a perimeter of the air permeable membrane is sealed to the fixed portion.
7. The transducer assembly of claim 6 wherein the perimeter of the air permeable membrane is sealed to a top side of the fixed portion of the surround, and a bottom side of the fixed portion of the surround directly below the perimeter is sealed to the frame.
8. The transducer assembly of claim 1 wherein the barometric vent comprises a triangular shape, a rectangular shape, a square shape, an elliptical shape or a racetrack shape.
9. The transducer assembly of claim 1 wherein the movable portion comprises a rolled portion that surrounds a piston portion that is operable to vibrate to generate sound.
10. The transducer assembly of claim 1 wherein the movable portion is symmetrical about two axes.
11. A portable electronic device comprising: an enclosure having a top cover and a bottom cover that define an interior chamber and an acoustic port to the interior chamber; a frame positioned within the interior chamber and sealed to the top cover or the bottom cover; a surround comprising a perimeter portion sealed to the frame and a movable portion operable to generate sound that

divides the interior chamber into a front volume chamber coupling a first side of the movable portion to the acoustic port and a back volume chamber coupled to a second side of the movable portion; and a barometric vent comprising an air permeable membrane sealed to the perimeter portion to vent a pressure between the front volume chamber and the back volume chamber.

12. The portable electronic device of claim 11 wherein the perimeter portion comprises a projection extending radially outward from the movable portion and the barometric vent is coupled to the projection.

13. The portable electronic device of claim 12 wherein the movable portion comprises a rectangular shape and the projection extends radially outward from a corner of the rectangular shape.

14. The portable electronic device of claim 12 wherein the movable portion comprises a hexagonal shape and the projection extends radially outward from a side of the hexagonal shape.

15. The portable electronic device of claim 12 wherein the projection extends radially outward from less than an entire perimeter of the movable portion.

16. The portable electronic device of claim 11 wherein the barometric vent comprises an air permeable membrane covering an opening through the perimeter portion of the surround and the frame, and a perimeter of the air permeable membrane is sealed to the perimeter portion of the surround.

17. The portable electronic device of claim 16 wherein the perimeter of the air permeable membrane is sealed to a top side of the perimeter portion of the surround, and a bottom side of the perimeter portion of the surround directly below the perimeter of the air permeable membrane is sealed to the frame.

18. The portable electronic device of claim 11 wherein the barometric vent comprises a triangular shape, a rectangular shape, a square shape, an elliptical shape or a racetrack shape.

19. The portable electronic device of claim 11 wherein the movable portion comprises a rolled portion that surrounds a piston portion that is operable to vibrate to generate sound.

20. The portable electronic device of claim 11 wherein the movable portion is symmetrical about two axes.
